Expt 4 Design of Low Pass Filter using Windowing Techniques

4..1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

CODE:

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect');
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');
%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');
figure;
```

freqz(b_hamm,1);

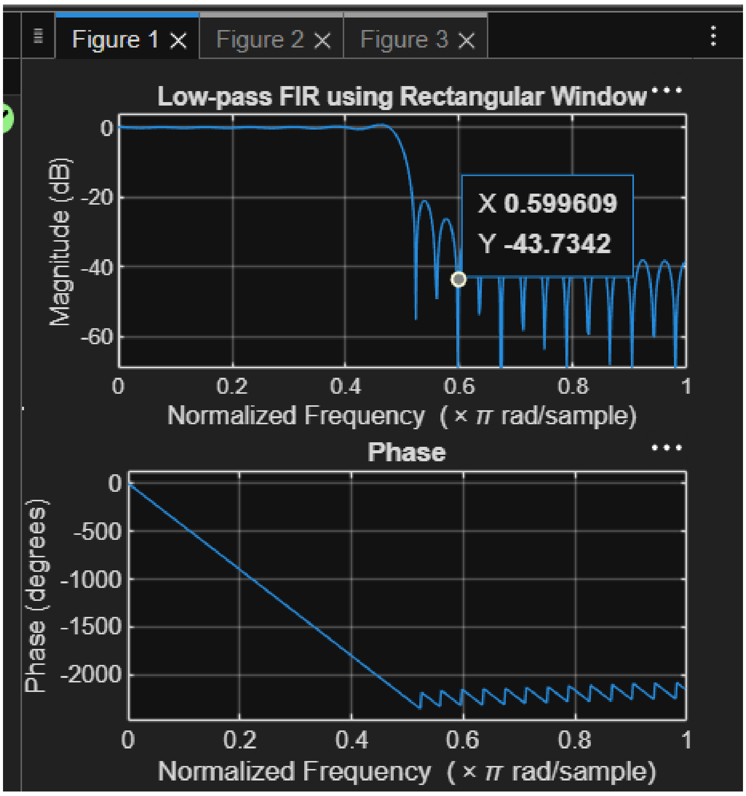
title('Low-pass FIR using Hamming Window');

figure;

freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');

OUTPUT;



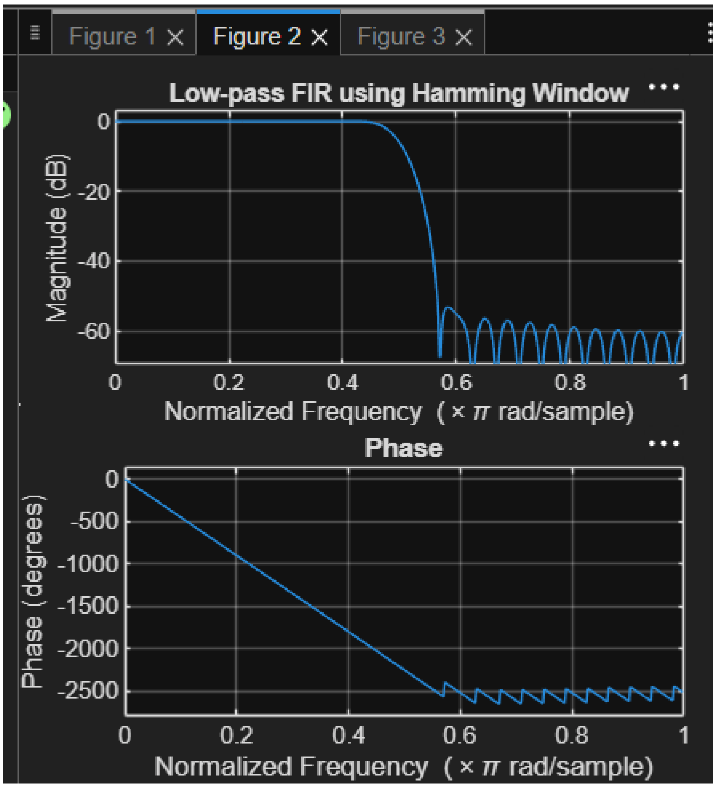


Figure 1
Figure 2
Figure 3
Low-pass FIR using Hamming Window
Magnitude (dB)
0
-20
-40
-60
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-500
-1000
-1500
-2000
-2500
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)

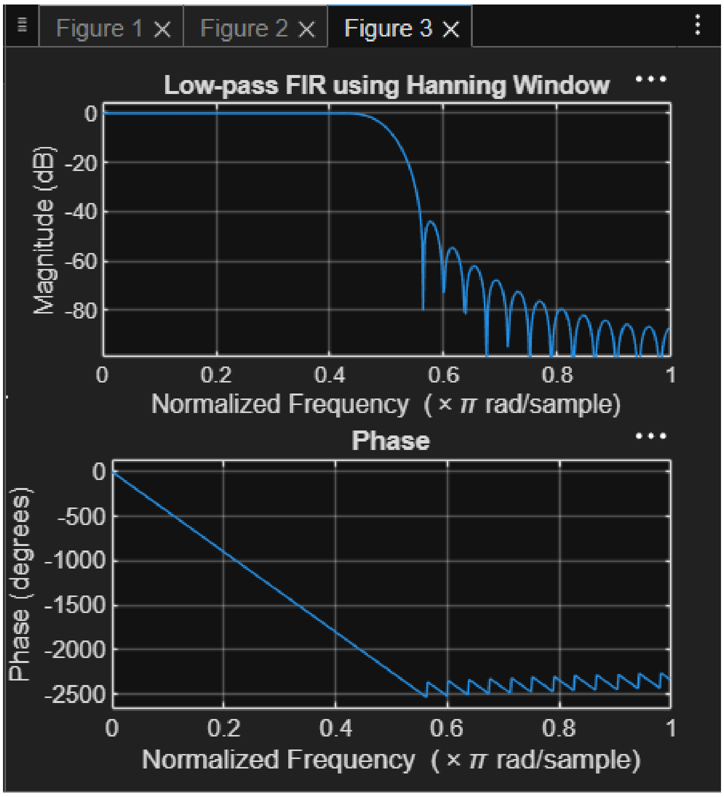


Figure 1
Figure 2
Figure 3
Low-pass FIR using Hanning Window
Magnitude (dB)
0
-20
-40
-60
-80
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-500
-1000
-1500
-2000
-2500
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)